universe, by and large, became transparent. Prior to the era of electron capture, the universe was filled with a dense plasma of electrically charged particles—some with positive charges like nuclei and others with negative charges, like electrons. Photons, which interact only with electrically charged objects, were bumped and jostled incessantly by the thick bath of charged particles, traversing hardly any distance before being deflected or absorbed. The charged-particle barrier to the free motion of photons would have made the universe appear almost completely opaque, much like what you may have experienced in a dense morning fog or a blinding, gusty snowstorm. But when negatively charged electrons were brought into orbit around positively charged nuclei, yielding electrically neutral atoms, the charged obstructions disappeared and the dense fog lifted. From that time onward, photons from the big bang have traveled unhindered and the full expanse of the universe gradually came into view.

About a billion years later, with the universe having substantially calmed down from its frenetic beginnings, galaxies, stars, and ultimately planets began to emerge as gravitationally bound clumps of the primordial elements. Today, some 15 billion or so years after the bang, we can marvel at both the magnificence of the cosmos and at our collective ability to have pieced together a reasonable and experimentally testable theory of cosmic origin.

But how much faith should we really have in the big bang theory?

Putting the Big Bang to the Test

By looking out into the universe with their most powerful telescopes, astronomers can see light that was emitted from galaxies and quasars just a few billion years after the big bang. This allows them to verify the expansion of the universe predicted by the big bang theory back to this early phase of the universe, and everything checks out to a “T.” To test the theory to yet earlier times, physicists and astronomers must make use of more indirect methods. One of the most refined approaches involves something known as cosmic background radiation.

If you’ve ever felt a bicycle tire after vigorously pumping it full of air, you know that it is warm to the touch. This is because when things are compressed they heat up—this is the principle, for example, behind pressure cookers, in which air is tightly compressed within a sealed pot